

MCIT — Digital Egypt Pioneers Initiative (DEPI) × eYouth

Graduation Project — Executive Summary

Skin Disease Diagnosis System

**Multimodal Deep Learning for Dermoscopic Lesion
Classification with Symptom-Based Reasoning**

Team

Yassin Taher — ML Engineer | Rawda Amr — Data Engineer
Mariam Akram — Computer Vision Engineer | Fady Mamdouh — Deployment Engineer

Supervised by Eng. Mahmoud Talaat

Project Idea

The project delivers an AI-powered diagnostic support tool that classifies dermoscopic skin lesion images into one of seven diagnostic categories, while also incorporating patient clinical context (age, sex, and lesion location) rather than relying on the image alone. The output is served through a deployed web service that returns a diagnosis, per-class confidence scores, a clinically actionable risk tier, and an independent ABCDE symptom check — mirroring how a dermatologist reasons about a case using both visual and contextual evidence.

Problem Statement

Dermatology faces a shortage of specialists relative to demand, so patients often wait weeks for an in-person evaluation of a suspicious lesion. Image-only classifiers can also be misled, since lesions that look visually similar can carry very different malignancy risk depending on a patient's age, sex, and lesion site. There is a need for a tool that fuses image and clinical evidence and clearly communicates risk level, to shorten time-to-triage and help prioritize urgent cases for both patients and clinicians.

Proposed Solution

A three-stage pipeline was designed:

- **Capture** — the user uploads a dermoscopic photo along with age, sex, and lesion location.
- **Fuse & Classify** — a multimodal deep learning network fuses image features with encoded clinical data to predict one of seven diagnostic classes.
- **Risk-Tier & Explain** — the prediction is mapped to a risk tier (Critical / High / Moderate / Low), paired with per-class confidence scores, Grad-CAM visual explanations, and an independent, rule-based ABCDE symptom checklist shown separately, so it never silently overrides the model's output.

The system was deployed as a FastAPI backend with a drag-and-drop web GUI, intended to serve teledermatology platforms (instant pre-screening), dermatologists and GPs (a second-opinion tool with visual transparency), and patients (fast, low-cost initial guidance).

Dataset

- **Source:** HAM10000 (ISIC 2018 Task 3) — a public dermoscopic image archive paired with patient metadata (age, sex, lesion localization), obtained from the official ISIC challenge repository plus a Kaggle-hosted metadata file.
- **Classes (7):** Melanoma (MEL), Melanocytic Nevus (NV), Basal Cell Carcinoma (BCC), Actinic Keratosis (AKIEC), Benign Keratosis (BKL), Dermatofibroma (DF), Vascular Lesion (VASC).
- **Size and split:** 10,015 raw images, split 60/20/20 into train / validation / test (6,009 / 2,003 / 2,003) using a stratified split, with an explicit check confirming zero image overlap between splits.
- **Key characteristic — severe class imbalance:** NV alone made up over half of the raw dataset (~6,700 images), while DF had under 100 images — roughly a 58:1 imbalance ratio, which directly motivated the class-balancing step described below.

Models / Algorithms Used

- **Image branch — EfficientNet-B4:** an ImageNet-pretrained CNN backbone (loaded via the timm library), chosen as a strong, parameter-efficient architecture well suited to dermoscopic image classification, used through transfer learning rather than training from scratch.
- **Clinical branch — MLP encoder:** a small network ($18 \rightarrow 64 \rightarrow 32$, ReLU + Dropout 0.2) that projects the structured clinical vector (age, sex, one-hot lesion location) into a 32-dimensional embedding.
- **Fusion & classification head:** the image and clinical embeddings are concatenated and passed through Linear $\rightarrow$ ReLU $\rightarrow$ Dropout 0.3 $\rightarrow$ Linear, producing logits over the 7 classes (softmax output).

This two-branch multimodal design was chosen specifically because image-only models cannot capture the way age, sex, and body site shift malignancy risk — fusing both modalities was the core hypothesis behind the project.

Preprocessing Techniques

- **Missing value handling:** age imputed with the training-set median; missing sex/localization set to an explicit "unknown" category.
- **Clinical feature engineering:** age z-score normalized using training-set mean/std; sex mapped to a numeric encoding (male = 1.0, female = 0.0, unknown = 0.5); lesion localization one-hot encoded across all anatomical sites plus an "unknown" bucket, producing an 18-dimensional clinical feature vector.
- **Image preprocessing:** resize to 224×224 with ImageNet mean/std normalization, applied to all splits.
- **Data augmentation (training images only):** random horizontal/vertical flip, random rotation ($\pm 20^\circ$), random resized crop (0.8–1.0 scale), random affine ($\pm 10\%$ translate, up to 10° shear), color jitter (brightness/contrast ± 0.2), and random erasing ($p = 0.2$) to simulate hair/occlusion artifacts. Validation and test images received only resize + normalization.
- **Class balancing:** minority classes were oversampled (image-level resampling, all majority-class images retained) up to the majority-class count of 4,023 images per class, combined with a WeightedRandomSampler during training.

Techniques and Technologies Used

- Transfer Learning — ImageNet-pretrained EfficientNet-B4 backbone fine-tuned end-to-end.
- Multimodal fusion learning — late fusion of CNN image features with an MLP-encoded clinical vector.
- Explainable AI — Grad-CAM heatmaps to visualize which image regions drove each prediction.
- Test-Time Augmentation (TTA) — predictions averaged over three views (original, horizontal-flip, vertical-flip) at inference.
- Mixed-precision (AMP) training for GPU efficiency.
- Rule-based clinical logic — an independent ABCDE (Asymmetry, Border, Color, Diameter, Evolving) symptom scoring system, kept deliberately separate from the model's prediction pipeline.
- Deployment stack — FastAPI backend serving a /predict endpoint, an HTML/JS drag-and-drop GUI, and ngrok tunneling for public access during development.

- Tools — Python, PyTorch, timm, Pandas, NumPy, scikit-learn, OpenCV, FastAPI, Google Colab, CUDA.

Training and Evaluation Methodology

- **Dataset splitting:** stratified 60/20/20 train/validation/test split with verified zero image leakage across splits.
- **Validation strategy:** no k-fold cross-validation was used; a single stratified validation split was monitored each epoch, with the best checkpoint saved whenever validation macro F1 improved.
- **Training setup:** up to 15 epochs, early stopping with a patience of 5 epochs (monitored on validation macro F1), batch size 32, mixed-precision training, and a WeightedRandomSampler over the balanced training set.
- **Evaluation metrics:** Accuracy, Macro F1-score (primary metric given class imbalance), Weighted F1-score, and per-class Precision / Recall / F1 via a full classification report and confusion matrix on the held-out test set.

Experiments Conducted

Two full training configurations were run on the same architecture and compared:

Experiment	Optimizer	Loss	Scheduler	Best Val Macro F1
Baseline (selected)	Adam (lr 1e-4)	Cross-Entropy	ReduceLROnPlateau	0.8254
Improved	AdamW (lr 1e-4, wd 1e-4)	Focal Loss ($\gamma=2$)	Cosine Annealing	0.7874

The Baseline configuration (plain cross-entropy + Adam + class-balanced oversampling) outperformed the Focal Loss + AdamW + Cosine Annealing configuration, and was selected as the final model. The added focal-loss weighting introduced variance without a clear macro-F1 gain within the 15-epoch training budget on an already-balanced training set.

Results

Final test-set performance (2,003 held-out images):

- Test Accuracy: 86.9%
- Macro F1-score: 0.786
- Weighted F1-score: 0.870

Per-Class Performance

Class	Precision	Recall	F1-Score	Support
MEL	0.65	0.63	0.64	223

Class	Precision	Recall	F1-Score	Support
NV	0.92	0.96	0.94	1,341
BCC	0.78	0.81	0.79	103
AKIEC	0.76	0.63	0.69	65
BKL	0.82	0.68	0.74	220
DF	0.89	0.74	0.81	23
VASC	0.92	0.86	0.89	28

Key Findings

- NV (the majority, benign class) was classified most reliably (0.94 F1), and rare classes (DF, VASC) benefited clearly from oversampling (0.81–0.89 F1).
- MEL (melanoma), the clinically most dangerous class, remained the hardest to classify (0.64 F1), most often confused with NV (69 of 223 melanoma cases misclassified as NV) — a clinically important gap, consistent with the difficulty dermatologists themselves face distinguishing these classes from a single image.
- Grad-CAM visualizations confirmed the model's attention consistently localized to the lesion itself rather than background artifacts.

For reference, this multimodal result (86.9% accuracy, 0.786 macro F1) compares favorably against a reported HAM10000 baseline single-image CNN (~85% top-1 accuracy, no clinical fusion) and typical image-only ResNet/DenseNet transfer-learning studies on this dataset (macro F1 typically 0.65–0.75).

Conclusion

The project demonstrates that fusing dermoscopic image features with structured clinical metadata (age, sex, lesion site) via a two-branch EfficientNet-B4 + MLP architecture produces a more context-aware, clinically informative classifier than an image-only baseline, while effectively handling severe class imbalance (58:1) through oversampling and weighted sampling. The system was carried through to a working deployment — a FastAPI inference service with test-time augmentation, a web GUI, Grad-CAM explainability, and a risk-tiered output paired with an independent ABCDE symptom check.

Main limitation: melanoma (MEL), the most clinically critical class, still lags at 0.64 F1 and is frequently confused with the benign NV class — meaning the system is not yet reliable enough to replace specialist evaluation and should be positioned strictly as a preliminary triage aid.

Future work identified by the team includes lesion segmentation prior to classification (to reduce background noise), calibrated confidence via temperature scaling, a medical chatbot for patient guidance, native mobile deployment, and exploring hybrid heads (EfficientNet features combined with XGBoost, Random Forest, or SVM) to further lift minority-class and melanoma recall.